

PHY 456 Tut 01:

Overview of Spherical Harmonics and probability flux.

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Probability current $\vec{j}$ on wavefunction $\psi(\vec{r}, t)$ or ψ^* .

$$\text{TDS E: } i\hbar \frac{\partial \psi}{\partial t} = H\psi = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right] \psi$$

with $P = \psi^* \psi$.

By chain rule,

$$\begin{aligned} \frac{\partial P}{\partial t} &= \psi^* \dot{\psi} + \dot{\psi}^* \psi \\ &= \frac{1}{i\hbar} \left[\psi^* \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi - \psi \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi^* \right] \\ &= \frac{i\hbar}{2m} \left[\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^* \right] \end{aligned}$$

by conjugate $(-i\hbar)^*$
potentials cancel since $V^* = V$.

For $V = A + iB$, this corresponds to a diffusion term of the amplitude of ψ , system loses energy over time.

Now: relate $\dot{P}$ to the divergence of a flux:

$$\rightarrow \frac{i\hbar}{2m} \left[\psi^* \partial_i^2 \psi - \psi \partial_i^2 \psi^* \right]$$

achieve P -current!

$$= \partial_i j_i \quad \text{for some current}$$
$$\vec{j} = \hat{e}_n j_n.$$

$$= \frac{i\hbar}{2m} \left[\psi^* \partial_i^2 \psi + \underbrace{\partial_i \psi^* \partial_i \psi}_{\text{invoke product rule}} - \psi \partial_i^2 \psi^* - \underbrace{\partial_i \psi^* \partial_i \psi}_{\text{invoke product rule}} \right]$$

invoke product rule:

$$= \frac{i\hbar}{2m} \left[\partial_i (\psi^* \partial_i \psi) - \partial_i (\psi \partial_i \psi^*) \right]$$

$$= \frac{i\hbar}{2m} \nabla \cdot (\psi^* \nabla \psi - \psi \nabla \psi^*)$$

$$\Rightarrow \frac{\partial P}{\partial t} + \nabla \cdot \vec{j} = 0$$

$$\text{with } \vec{j} = -\frac{i\hbar}{2m} (\psi^* \nabla \psi - \psi \nabla \psi^*)$$

is the continuity equation for probability.

Next: polynomial review.

• Legendre Polynomials:

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2 - 1)^l$$

for $l \in \mathbb{Z}^{\geq 0}$

which form a complete subset of functions with finitely many discontinuities on $[-1, 1]$.

The orthonormal relation:

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{nm}$$

And any $f(x) = \sum_{l=0}^{\infty} a_l P_l(x)$

with $a_l = \frac{2l+1}{2} \int_{-1}^1 f(x) P_l(x) dx$

Similar to a Fourier transform.

• Spherical Harmonics:

$$Y_l^m(\theta, \varphi) = N e^{im\varphi} P_l^m(\cos\theta)$$

↑
"tensor"/matrix

↑
normalization

↑
Associated Legendre Polynomials

where $P_l^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m}{dx^m} P_l(x)$

- Plane waves, Bessel Functions:

($z = r \cos \theta$ in spherical coordinates)

Plane wave

$$e^{ikz} = e^{ikr \cos \theta} \\ = \sum_{l=0}^{\infty} i^l (2l+1) j_l(kr) P_l(\cos \theta)$$

where

$$j_l(x) = (-1)^l x^l \left(\frac{1}{x} \frac{d}{dx} \right)^l \frac{\sin x}{x}$$

$\uparrow$ Spherical Bessel Function $\uparrow$ sinc(x)

Since most scattering is done with initial plane waves.